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Q. Differentiate between horizontal and vertical gene transfer?

Ans: Horizontal Transfer of genes occurs between different organisms while Vertical Transfer of gene occur top to bottom like from parents to offspring. Horizontal Transfer of genes between different organisms is a relatively slow process that leads to acquisition of new traits.

Q. Enlist Edit operation?

Ans: Edit operations are insertion, deletion, and substitution of one symbol from another.

Q. What do you know about READSEQ?

Ans: READSEQ is a useful sequence converter developed by D.G.Gilbert at Indiana University, USA. Recognizes DNA or Protein sequence file and interconvert them.

Q. How can you define an Algorithm?

Ans: Algorithm is a sequence of instruction that must be accomplish to solve a well formulated problem. It can translate problems in the form of input to out put.

Q. Define variable and array?

Ans: Variable:

It is written as "X" or the total. It is a term in which we sign some numerical values to variables, and this process can be change during process of algorithm.

Array:

It is an element that is an ordered collection of some specific numbers or n variables. So n variables represents by (a₁,a₂,a₃a_n).

Q. Comparative proteomics and its advantages?

Ans: Genes encode proteins and these proteins perform actual life functions, we can have the genome and get the translated protein from it and by comparing those translated proteins with one another, we can say it's a Comparative Proteomics study. The collection of the protein sequences that are encoded by the genome, makes up the proteome of that individual. From a Genome sequence, we can get the predicted genes out of it, then we do translations which gives us the proteome of that individual. We can use those proteins and get some Database searches (we can find the homologous proteins, whereby we can predict their functions and roles also do the comparison of proteome by themselves as in self comparison so that will help us finding the paralogs within that individual. We can also do the comparison of proteins between different organisms and we can also find the clusters of co related protein in term of their sequences and functions.

Q. Explains the regions of DNA?

Ans: DNA has these basic regions. Coding region and regulatory region.

Regulatory region do not take part in the translation of protein have 2 types,

Upstream region: 5' end translated Downstream region: 3' end translated

Exon: Exons are coding areas. exons are very much conserved. Exons can be termed as DNA bases which are translated into mRNA.

Introns: introns are non-coding areas. Introns are less conserved which means that their sequence changes very frequently over time. Introns are also DNA bases that are found in between exons.

Q. Define, alignment... match... mismatch... insertion... deletion?

Ans: Alignment; it is a arrangement in a straight line or in correct relative position.

The **alignment** of the strings **v** (of *n* characters) and **w** (of *m* characters, with *m* not necessarily the same as *n*) is a **two-row matrix** such that the first row contains the characters of **v** in order while the second row contains the characters of **w** in order, where spaces may be interspersed throughout the strings in different places As a result, the characters in each string appear in order.

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Match: These are the same characters present between two sequences when we align two sequences.

Mismatch: These are the different characters present between two sequences when we align two sequences.

Insertion: Insertion are the mutation in which extra base pairs are inserted or add into a new place in the DNA.

Deletion: deletions are the mutations in which a section of DNA is lost or deleted.

Q. Write a short note on mitochondrial DNA and chloroplast DNA?

Ans: **Mitochondrial DNA** (both in animals and plants), **Chloroplasts DNA** (in plants), these are membrane-bound organelles and they may be present in hundreds to thousands of copies (so there is also multiple copies of these genomes). Mitochondrion is the site for respiration whereas chloroplast is the site for photosynthesis. Their DNA's can be labeled as mtDNA or cpDNA respectively. Plasmids, yeast, Transposons, Viral genomes and retroviruses are other examples of organellar DNA.

Q. What do you know about the collinear arrangement of DNA?

Ans: In general it is the arrangement of one sequence in the same linear order as another sequence. In the case of DNA it is the arrangement of genes on chromosome in a same order or correspondence between the linear order of nucleotides codons in the DNA.

Q. Write the names of six model organisms?

Ans: *E. coli* – bacteria
• *S. cerevisiae* – yeast
• *C. elegans* – worm
• *D. melanogaster* – fly
• *Daniorerio* – zebrafish
• *Mus musculus* - mouse
• *Homo sapiens* – you and me
• *Arabidopsis* - plant

Q. Write the BIF application in molecular medicine and biotechnology?

Ans: **Molecular Medicine**

The Human Genome Project Data Base search that there are 3000-4000 hereditary disease including Cystic Fibrosis and Huntingtons disease. Response to an environmental stress causes cancers, heart disease, diabetes.

Biotechnology:

it is very vast field, in which we use microorganism to form different products for the beneficial of mankind.

For example:

Corynebacterium glutamicum ; it produce the lysine.

Xanthomonas campestris ; it produce xanthium gum.

Lactococcus lactis. It is use to produce butter and cheese.

Archaeoglobus fulgidus and *Thermotoga maritima* ;

Q. what do you mean by highly conserved region in an alignment?

Ans: Exons are very much conserved in nature. These are coding areas. Exons can be termed as DNA bases which are translated into mRNA.

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But in the case of alignment highly conserved region is that which is similar in both sequence of two organisms and also specifies for a particular function. For example Homo box gene involved in embryonic development present in wide range of organisms.

Q. What do you know about local alignment problem?

Ans. Local Alignment Problem:

When Biological significant sequence are present in certain parts of DNA fragments and not present in the other DNA sequence, the biologists have maximize the alignment score $s(v_i \dots v_{i'} \text{ prim upto } w_j \dots w_{j'})$. These matrix score spread over all substrings $v_i \dots v_{i'}$ of v and $w_j \dots w_{j'}$ of w , known as Local Alignment Problem. LAP only concern the specific part of the DNA not extend over entire length as in Global Alignment Problem.

So in LAP our target is to find the best local alignment b/w two strings.

Local Alignment Problem:

Input: string “v” and “w” with a scoring matrix δ .

Output: Substring of v and w whose global alignment, (as defined by δ), is maximal among all global alignment of all substrings of v and w .

It only covers parts of the sequences to be aligned.

- Smith-Waterman algorithm finds the best local alignment between two sequences

Q. Write the purpose of submission data base and its caution?

Ans. Submission of sequence database:

Sequences are submitted to the databases in order to share them with the scientific community (sometimes they are also required by the Publication and funding agencies to submit them). Generally sequences are submitted at the time of publication and are reviewed by peers. Biological databases in general store biological data and their main goals are

- Data storage
- Information retrieval
- Knowledge discovery
- Nucleotide Sequence Databases

Caution:

It is important to ensure that sequence files do not contain any special characters because sometimes the control characters can also be incorporated into or normal sequences, which can then mess-up the downstream analysis or data-transfer. So, there is an issue of how to put the ambiguous nucleotides or amino acids in the sequences (because at some places you are not sure whether it is ‘A’ or ‘T’ or ‘G’ or ‘C’ and you are restricted to put a single letter). So, there is an organization known as International Union of Biochemistry (IUB), it has established some standard codes to represent those ambiguous bases or amino acids. For example, here we see that G, A, T or C are just Guanine, Adenine, Thymine and Cytosine respectively. If we see R, it can be either A or G and the word is derived from the group they are coming from i.e. the purines. We see Y that is the pyrimidine, it can be C or T. M stands for if they are having some amine group / amino group in them, K is if they have Keto group i.e. G or T. S is if they have strong interactions (3 hydrogen bonds) like C and G, who forms triple bonds. W is for weak interactions, A or Since H follows G in Alphabet so it’s everything except G, it can be A, C or T and similar procedure is followed for B, V, and D whereas N can be any base. Similarly, for amino acids, we have single letter codes i.e. from A to Z. And we can see in the figure on the left that some letters are missing. There are four amino acids that are starting with G, but we gave

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that G letter to Glycine and for rest of them, we might use some other letters like Glutamic acid is represented as E. Y stands for Tyrosine (down below) and X can be any amino acid like N (in case of the nucleotide sequences).

Q. Write the stages of genome annotation? (3)

Ans: There are two important steps of genome annotation:

1. Structural annotation: Identification of gene features like;

- Promoters
- Terminators
- Shine-Dalgarno sites
- DNA motifs (patterns of nucleotides within the genes)
- Co-transcription units

• Operons in microbes. In micro-organisms, lots of genes are transcribed together known as operons. (MAGPIE database is often use).

2. Functional annotation: Attaching biological information to genes like;

- Biological function
- Biochemical function
- Gene expression
- Regulation and interactions. (GENEQUIZ is often use)

Three steps in genome annotation:

1. Find features associated with protein-coding genes (e.g. tRNA, rRNA, snRNA, SINE/LINE, miRNA precursors)
2. Build models for protein-coding genes, including exons, coding regions, regulatory regions
3. Associate biologically relevant information with the protein coding genome features and genes.

Q. What is largest common subsequence and its purpose?

Ans: **Largest Common Subsequences (LCS)**

The simplest form of a sequence similarity analysis is the Longest Common Subsequence (LCS) problem, where we eliminate the operation of substitution and allow only insertions and deletions. A subsequence of a string v is simply an (ordered) sequence of characters (not necessarily consecutive) from v . Typically there are many common subsequences between two strings v and w , but how to find the longest one. $s(v, w)$ be the length of the longest common subsequence of v and w edit distance between v and w under the assumption that only insertions and deletions are allowed.

$$d(v, w) = n + m - 2s(v, w)$$

This corresponds to the minimum number of insertions and deletions needed to transform v into w .

Longest Common Subsequence Problem:

Find the longest subsequence common to two strings

Input: Two strings, v and w

Output: The longest common subsequence of v and w

Q. What are the factors on which the genome of different organisms can be compared?

Ans: We can compare the genomes of different organisms like we do comparison of gene number, gene content and gene location in both prokaryotic and eukaryotic groups of organisms.

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Q. what are the main classes of biological database with example?

Ans: Biological databases can be classified as

- **Primary databases**
- **Secondary databases**
- **Specialized Databases**
- **Nucleotide Sequence Databases**

It can also be classified as,

Nucleotide databases, Protein databases, RNA databases, Genome databases & Expression databases.

Nucleotide Sequence databases

Nucleotide Sequence Databases store DNA and cDNA or EST sequences.

Protein Databases

Protein databases store protein data which may include the following:

- Protein sequences
- Motif (patterns of amino acids)
- Protein Structure
- Structure alignments (aligned structures)

Genome & Organism Specific Databases

First attempt to sequence free living organism was launched in late 1990's (Blattner et al. 1997) and Viruses had already been sequenced (Fleischmann et al. 1995).

AceDB (AC. *Elegans* DataBase) was the first genome database for genome sequences was developed in 1989 and was established by **Richard durbin** and **Thierry-Miegi**.

Gene Expression Databases:

Gene Expression Databases are the public repository for the archiving and distribution of gene expression data submitted by the scientific community. Genes are expressed into mRNA, and whenever we talk about gene expression, we generally mean the mRNA sequences so we can normally get those mRNA for submission and further analysis.

Q. What are the criteria of organism selection for sequence?

Ans: Selection Criteria:

There are different criteria for the selections of the organisms which are then send to the genome sequencing projects. Following are the criteria for selection:

- They had been subjected to a detailed biological analysis/ extensive studies and thus were model organisms.
- They might be important human pathogens (so it's important for us to study its genome).
- They were of phylogenetic interest.
- Sequences were annotated as they were sequenced.

Q. What is VNTR?

Ans: VNTR is an example of mini satellite. They are 15 bases long in array of several hundred to thousands kb. They are typically present in euchromatin region. Used to identify human individuals in forensics.

Q. Write a note GEO? (5)

Ans: Gene Expression Omnibus (GEO):

Genes are expressed into mRNA, and whenever we talk about gene expression, we generally mean the mRNA sequences so we can normally get those mRNA from techniques like microarray and another famous technique nowadays which is being established is known as

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RNAseq. And microarray data and RNAseq can be classified into Gene Expression Data which is stored in Gene Expression Databases. Basically, Gene Expression Databases are the public repository for the archiving and distribution of gene expression data submitted by the scientific community. The gene expression data are MIAME compliant data (where MIAME is Minimum Information About a Microarray Experiment and can be searched on this link:

<http://www.mged.org/Workgroups/MIAME/miame.html>).

Whenever we need to submit a microarray experiment, we need to give descriptions like how exactly the samples were prepared, normalization methods which we have used and other counts and normalized files so while keeping in view these datasets have different records in it. Gene Expression Omnibus is convenient for deposition of gene expression data, as required by funding agencies and journals and it's also a curated resource for gene expression data where we can do

Browsing, querying, analysis and retrieval of the data. The webpage of GEO which is Gene Expression Omnibus running under NCBI which are having different datasets, has expression profiles where we can see the change in expression of genes across different treatments and we can also analyze this expression data. There is a tool called as GEO2R, we can use BLAST in it.

Q. Explain exon prediction approach?

Ans: Exons can be predicted within ORF from the information gathered about

- splice junctions

Approach:

For each segment [acceptor, donor], we get three scores (coding potential, donor score, acceptor score)

Various possibilities

- all three scores are high – probably true exon.
- all three scores are low – probably not a real exon.
- all in the middle -- ?.

Some score are high and some are low.?

we can get the evidence by the help of that information which we gathered from those splice sites.

Prediction:

- Collect a set of exons and non-exon
 - Score them using our scoring schemes
 - Plot them as follows
- draw a separate line between exon and non-exon.

Linear Discriminate Analysis: (LDA)

- linear discriminate analysis (LDA) finds an optimal plane surface that best separates points that belong to two classes.
- For example, if there are ten true exons and ten introns, and two feature.
- These samples could be represented by 20 points in a two
- LDA would compute a straight line through the space that can best separate the two classes with the minimal classification error.

Q. Define the motif problem?

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Ans: In this problem we have given a set of DNA sequences, we have to find a set of l-mers, one from each sequence, that maximizes the consensus score.

Q. What is the split gene?

Ans: In 1977 the concept of split genes introduced by Phillip Sharp and Richard Roberts work on Adenovirus, These genes are not continuously stretched thought the DNA rather there are small pieces of another type of DNA that is called Junk DNA that do not code for any protein. Split genes are analogous to a magazine article- page 1, 13, 43, 51, 74, 80, and 91, with pages of advertising appearing in between. Junk DNA represents “advertising” that separates exons.

Q. STS stands for?

Ans: it stand for sequence tag sites.

Q. Define partial consensus?

Ans: $s = (s_1, s_2, \dots, s_t)$, define the partial consensus score.

Q. What is waardenburg's syndrome?

Ans: Named after a Dutch biologist which proposed that it is genetic disorder in the human loss of hearing and some pigmentry abnormalities. It leads to two different eye color in one individual. He says that the individual with the pigmented abnormalities also have hearing problem.

Q. GeneBank, EMBL, DDBJ?

Ans: **GeneBank** sequence format is found to be in the Gene Bank Database which is a kind of standard format and other formats are pretty similar to it. A sequence file in Gene Bank format can contain several sequences. One sequence starts with a line containing the word LOCUS and a number of annotation lines. The start of the sequence is marked by a line containing "ORIGIN" and the end of the sequence is marked by two slashes ("/").

EMBL format is similar to that of Gene Bank Format. European Molecular biology Lab (**EMBL** established 1980). Here, we have ID, accession number (AC) and descriptions (DE). Sequence actually starts from where the word 'SQ' is there. Finally, the sequence ends with doubles slashes ("/") same as in Gene Bank format.

DNA databank of Japan (DDBJ):- established in Mishima japan (1984). Genebank, DDBJ and EMBL joined together in **International Nucleotide Sequence Database Collaboration (INSDC)**. Databases store sequences in specified formats. Different software needs sequences in different formats. Genebank, DDBJ and EMBL has similar formats.

Q. What is the basic purpose of IUB?

Ans: IUB (International Union of Biochemistry) has established some standard codes to represent those ambiguous bases or amino acids. For example R is use for the purines Y for pyrimidin etc.

Q. Write a short note pre order algorithm?

Created By: CR. SABA ALWI Page 9

Ans: There are three basic search tree algorithms.

1. Next leave
2. All leaves
3. Preorder

Preorder is a type of recursive algorithm.

It assigns a number to vertices in a transversing order. Either in the left or right direction.

Q. What is effect when we delete 1 nucleotide, 2 or 3 nucleotides?

Ans: Sydney Branner and Francis Crick demonstrated that when 1 nucleotide is deleted the protein has very drastic effect, when 2 nucleotides are deleted protein also has drastic effect that transcribe from that gene and could not properly function, but when we delete 3 nucleotides consecutively then there would be a minor change as the result of this deletion.

Q. Write a note enzyme commission number with example?

Ans: Enzyme Commission (EC) numbers:

It is another scheme which was put forward by the Enzyme Commission (EC) that was working under the **IUBMB (International Union for Biochemistry and Molecular Biology)**. They say that the enzymes are classified on the basis of the reactions they catalyze and have a 4-digit scheme which is actually the enzyme commission number: **EC a.b.c.d**

'a' (first digit) informs that it is from one of the 6 classes of biochemical reactions (enzyme might be coming from one of these classes).

'b' (second digit) informs that it is from the group of substrate (the thing on which the enzyme attacks).

'c' (third digit) informs us that it is an acceptor molecule.

'd' (fourth digit) gives the details of biochemical reaction

For example,

tripeptidaminopeptidases

EC 3.4.11.4

Where **3** – tells us that it is a Hydrolase (use water to break substrate).

This **3.4** – tells us that it is a Hydrolase that acts on the peptide bonds.

The **3.4.11** – tells us that it is a Hydrolase that cleaves the amino terminal amino acids of polypeptide. While putting everything together, **EC 3.4.11.4** – it tells us that it is a Hydrolase that cleaves the amino terminal amino acids of a tri-peptide.

Q. How microbes use for climate change?

Ans: Increasing levels of carbon dioxide emission cause change in global climate. Study the genomes of those microbes is very useful that use carbon dioxide as their sole carbon source.

Q. Write the four cases of pattern finding?

Ans: Pattern Finding in a Genome:

— A **pattern** (keyword) is an ordered sequence of symbols .

— Symbols of the pattern and the searched text are chosen from a predetermined finite set, called an **alphabet** (Σ)

Four Cases of Pattern Finding:

— Look for a perfect match

— Allow errors due to substitutions

— Allow errors due to insertions-deletions (InDels).

Rank possible matches according to a weight function and keep matches above a certain threshold. For every pair, aligned together, we come up with the weight function and in the end we combine all those weight functions.

Q. Differentiate b/w SINES and LINES?

Ans: Non-LTR (Long Terminal Repeats) retrotransposon comprises of **SINES** and **LINES**.

SINES are Short Interspersed Nuclear Elements and are **80-300 bp** long and the example is

Alu repeats (found in humans). Whereas **LINES** (Long Interspersed Nuclear Elements) are **6-8 kb** long and are relatively longer than SINES.

Q. What are NF-KB binding site?

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Ans: These are Specific sequence located upstream of genes in *D.melanogaster*.

TCGGGGATTTC NF- κ B binding sites (nuclear factor kappa-light-chain-enhancer of activated B cells).it helps in activation of immunity gene in fruit fly.

Q. Write the two ways of sequence submission in NCBI?

Ans: NCBI has two options for sequence submission

BANKIt – It is responsible for the submission of small data set. Often use for simple sequences (not related with down-stream analysis) and annotations and can be submitted through web. which does not requires any advanced tools.

Sequin - For Complex sequences and annotations and is also good if we want to do some off-line submissions normally where we have our datasets which are huge ones and can be used in future with some advanced tools (for analysis) and graphical reports.

Q. Write about Donor and Acceptor splice site in splice junction?

Ans: There exists a weakly conserved sequence of eight nucleotides at the boundary of an exon and an intron (donor splice site) and a sequence of four nucleotides at the boundary of an intron and exon (acceptor splice site).

Q. Write power of DNA comparison?

Ans: The Power of DNA Sequence Comparison:

On Discovery of new gene there was no idea of functions. Scientist Find similarities between unknown genes with the genes of known function. Following are some evidences, *Newly discovered cancer-causing -sis oncogene matched with a normal gene involved in growth and development called platelet-derived growth factor. Oncogene v-sis is the simian sarcoma virus. Scientists became suspicious that cancer might be caused by a normal growth gene.

*Discovery of cystic fibrosis gene that involve abnormal secretions, and is diagnosed in children produce abnormally thick mucus that clogs the lungs. 10 million Americans are carriers of the cystic fibrosis gene. There is a 25% chance that the child will have cystic fibrosis In 1989, 1million nucleotides on the chromosome 7. The area around the cystic fibrosis gene was sequenced.

*Database of all known genes Similarities between a gene that had already been discovered, code for adenosine triphosphate (ATP) binding proteins. These proteins constitute the ion transport channel.

*The disease involves sweat secretions with abnormally high sodium content

Link between cancer-causing genes and normal growth genes and elucidating the nature of cystic fibrosis.

Q. What are tandem repeats and interspersed repeats?

Ans: Genome in the eukaryotic cell present in repeats, that can be separated on the bases of densities. These are of two types. Tandem repeats (array of repeats together), interspersed repeats (those repeats which are separated by normal genome followed by repeats i.e. normal genome is between those repeats).

Q. Stop and start codon and their purpose?

Ans: Start codon is **ATG**. In the process of transcription the RNA polymerase bind to the promoter region of gene start the transcription at these start codons.

The three “stop” codons, (**TAA**, **TAG**, and **TGA**). Transcription will be stop while arriving the stop codon.

Q. Write the function delete (y,L) with example?

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Ans: In Brute Force Algorithm we make a List of pairwise distances, L, and uses the function DELETE(y, L). It means the value of “y” will be deleted from the subset “L”.

Q. Write the purpose of genome analysis? 5

Ans: Genome Analysis:

In Genome Analysis we mainly perform the following tasks;

- **Sequencing and Assembly:** Since the sequencing is done in a way that whole genome is broken down into short fragments and once those fragments are sequenced, we need to put them together, this step in genome analysis is known as Assembly.

- **Gene prediction:** After we have assembled a finished genome, now we can go for the prediction of the genes where we can find the genes by using different patterns or features of those genes.

- **Repeat identification and masking out:** Once we assemble that genome, we try to find out the regions in which we have large number of repeats because assemblies jumble up where we have those repeats so we need to find those regions and it is one of the important task to go and look into those assemblies while keeping in mind those regions in which we have those repeats.

- **Looking for EST (Expressed Sequenced Tags) and cDNA (complementary DNA) sequences:**

EST and cDNA are basically originated from the DNA where the genes that are expressed are transcribed into mRNA which is then reversed transcribed back into cDNA. So by the help of cDNA, we can look into where those expressing regions are present in the genes that will give us the idea of the gene expression or the regions from where the mRNAs are made.

- **Expression analysis:** Once we have the idea about the regions of the gene in which we can have the gene expression then we can explore the quantification, i.e. how much those genes are being expressed.

- **Genome annotation:** In which we can find out similar functions performed by different genes)

- **Metabolic pathways and regulation studies:** Once those genes are expressed, their products interact with each-other and then they perform different metabolic roles in the shape of different metabolic pathways and networks.

- **Functional genomics:** Where we are actually looking into the different functions performed by different regions of the genome, that are under the control of different genes. and what exactly would be the effect of changes in those genes specifically if we want to study about the genes related to diseases.

- **Gene location/gene map identification:** Map the location of those genes on the chromosomes.

- **Comparative genomics:** In which we can take one genome and compare it with another genome, where we can find the comparative features; what is present in the first genome and not in the second one and the intersections between them, etc.

- **Identify clusters of functionally related genes:** Those genes they might be having similar structures, sequences and also performs similar functions, which can give us the idea about the evolution.

Q. Define ORF?

Ans: During the gene expression, both exons and introns are first transcribed into mRNA and then the introns are removed out by splicing, so the remaining structure is known as ORF (Open Reading Frames) which only consists of the exons.

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An ORF is a sequence of DNA that starts with start codon "ATG" (not always) and ends with any of the three termination codons (TAA, TAG, TGA)

Q. Define Gold Bug theory?

Ans: Gold bug by Adgar allan provide some clues of finding DNA motives, there are some coded information that is written in integers ";48" is use repeatidly in this coding. As "THE" word is mostly use in our stories.so he used "THE" word instead of ;48.

; for T

4 for H

8 for E

Q. Brute force and greedy approach?

Ans:

Q. PAM and PAM1 difference?

Q. PAM and PAM1 difference?

Created By: CR. SABA ALWI Page 13

Ans: pointed accepted mutation (PAM) and PAM1 unit is the amount of time in which average protein mutate to only 1% of its amino acids.

PAM1 is also a matrix in which there is a 1% difference in amino acid so it can define many alignments of extreme sequence within a protein.

Q. Define In frame hexamer?

Ans: In-Frame hexamer count: This approach discovered by Mark Borodovsky. It counts for frequencies of two adjacent codons present in a coding region i.e. frequencies of pairs of consecutive codon. It is more applicable in bacterial genome. Because of lack of intron and presence of conserved region on the starting of transcription. This approach not much applicable on eukaryotes, because of very complex genome pattern. In vertebrate the average size of exon is just 130 nucleotides. This size is much small to be consider because these small size exons do not have reliable peaks in likelihood ratio plots, while we are examining ORF. They do not differ from random fluctuation ion to be detectable.

Q. Differentiate satellite DNA, micro and mini DNA?

Ans: Satellite DNA:

Satellite DNA has following features:

- It may be one to several thousand bp long and it can also be present as million bases long.
- They are present near centromere and telomere. It can be classified as mini satellite and micro satellite.

Mini-satellite:

Mini-satellite features are as follows:

- They are 15 bases long in array of several hundred to thousands kb.
- They are typically present in euchromatin region.
- Example is VNTR and is used to identify human individuals in forensics.

Microsatellite:

Micro-satellite features are as follows:

- They are 2-6 bases long and can be in arrays of 10-100 bases.
- They are inherited to offspring.
- Mainly they are useful markers for genetic analysis and evolutionary studies (just like inprevious example)

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- They are found in telomeres TTAGGG (one example of six nucleotides segment).
- SSRs and STRs (Simple Segment Repeats and Short Tandem Repeats) are typical examples.

Q. Write down the mutable genes?

Ans: Asparagine (Asp), Serine (Ser) Glutamate (Glu), Aspartate (Asp) are **More mutable** while Cysteine (Cys), Tryptophan (Trp) are **least mutable**.

Q. write Edit distance? 5

Ans: Edit Distance

Edit distance between two strings as the minimum number of editing operations needed to transform one string into another, where the edit operations are insertion, deletion, and

Created By: CR. SABA ALWI Page 14

substitution of one symbol for another. It is often the case that the i th symbol in one sequence corresponds to a symbol at different position in other. It can cause the mutation in DNA. And also play a significant role in evolutionary process and DNA replication. Substitutions, insertions, and deletions of nucleotides, leads to “edited” DNA texts. Whether the i th symbol in one DNA sequence corresponds to the i th symbol in the other. Less edit distance better the solution of a problem.

Q. Write a note on uniprot?

Ans: Uniprot is a protein database it formed by International partnership between PIR, EBI and SIB Created UniProt, by unifying the PIR-PSD, Swiss-Prot, and TrEMBL databases.

Page of the uniprot consist of 3 different sections,

1. **PRO** (protein ontology): here we classify those protein on the bases of their functions. Different function has their different hierarichy.
2. **Pro class:** it is the region where we have the sequence.
3. **Pro link:** It provides us literatures information and most the search papers.